

Formula sheet

- Conversion from binary to decimal:

$$(a_n a_{n-1} \cdots a_1 a_0 . a_{-1} a_{-2} \cdots a_{-m})_2 \\ = a_n \times 2^n + a_{n-1} \times 2^{n-1} + \cdots + a_1 \times 2^1 + a_0 \times 2^0 + a_{-1} \times 2^{-1} + a_{-2} \times 2^{-2} + \cdots + a_{-m} \times 2^{-m}$$

- Taylor expansion (around $x = a$):

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

- Definitions of vector norms (for $\vec{x} \in \mathbb{R}^n$):

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}, \quad \|\vec{x}\|_1 = \sum_{i=1}^n |x_i|, \quad \|\vec{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

- Definitions of matrix norms (for $A \in \mathbb{R}^{n \times n}$):

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|, \quad \|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|, \quad \|A\|_2 = \max_{1 \leq i \leq n} \sigma_i$$

where σ_i 's are the singular values of A (eigenvalues of AA^T).

- Definitions of root finding iterations:

Fixed point iteration (to find $x^* = g(x^*)$): $x_{k+1} = g(x_k)$ for $K \geq 0$

Newton's iteration (to find $f(x^*) = 0$): $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ for $n \geq 1$

Secant iteration (to find $f(x^*) = 0$): $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$ for $k \geq 1$

- Summation identities:

$$\sum_{k=1}^{n-1} 1 = n-1, \quad \sum_{k=1}^{n-1} k = \frac{1}{2}n(n-1), \quad \sum_{k=1}^{n-1} k^2 = \frac{1}{6}n(n-1)(2n-1)$$

- Iterative methods for solving $Ax = b$ (where A_D, A_L, A_U , are the diagonal, lower-triangular and upper-triangular parts of A):

Jacobi iteration: $x^{n+1} = -A_D^{-1}(A_L + A_U)x^n + A_D^{-1}b$,

Gauss-Seidel iteration: $x^{n+1} = -(A_D + A_L)^{-1}A_Ux^n + (A_D + A_L)b$,

- Lagrange interpolation on data $\{(x_i, f_i)\}_{i=0}^n$

$$f(x) \approx p_n(x) = \sum_{i=0}^n f_i L_i(x), \quad L_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$$

- Hermite interpolating polynomial (given data $\{(x_i, f_i, f'_i)\}_{i=0}^n$):

$$H_n(x) = \sum_{i=0}^n f_i h_i(x) + \sum_{i=0}^n f'_i \tilde{h}_i(x)$$

where $h_i(x) = \left(1 - 2L'_i(x_i)(x - x_i)\right) (L_i(x))^2$ and $\tilde{h}_i(x) = (x - x_i) (L_i(x))^2$

- Quadrature rules for $\int_a^b f(x) dx$:

Rule	Truncation error	Degree of precision
Midpoint:	$\hat{I}_M(f) = (b - a)f\left(\frac{a+b}{2}\right)$	$\frac{(b-a)^3}{24} f''(\xi_0)$ 1
Trapezoid:	$\hat{I}_T(f) = \frac{b-a}{2} (f(a) + f(b))$	$-\frac{(b-a)^3}{12} f''(\xi_1)$ 1
Simpson's:	$\hat{I}_S(f) = \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right)$	$-\frac{(b-a)^5}{2880} f^{(4)}(\xi_2)$ 3
Composite trapezoid:	$\hat{I}_{CT}(f) = \frac{h}{2} [f_0 + 2f_1 + \dots + 2f_{n-1} + f_n]$	$O(h^2)$ where $h = \frac{b-a}{n}$ 1
Composite Simpson's:	$\hat{I}_{CS}(f) = \frac{h}{6} [f_0 + 4f_{1/2} + 2f_1 + \dots + 4f_{n-1/2} + f_n]$	$O(h^4)$ where $h = \frac{b-a}{n}$ 3
General:	$\hat{I}(f) = w_1 f_1 + w_2 f_2 + \dots + w_{n-1} f_{n-1} + w_n f_n$	— —

where w_i are the weights and $f_i := f(x_i)$

Note that if a quadrature rule has precision m it means that it will exactly integrate polynomials of degree m or less exactly.

$$\begin{aligned} \text{Midpoint rule:} \quad & \hat{I}_M(f) = (b - a)f\left(\frac{a+b}{2}\right) \\ \text{Trapezoidal rule:} \quad & \hat{I}_T(f) = \frac{b-a}{2} (f(a) + f(b)) \\ \text{Simpson's rule:} \quad & \hat{I}_S(f) = \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right) \end{aligned}$$

- Real form of the Fourier series

$$\begin{aligned} f(t) &= \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k \cos\left(k \frac{2\pi t}{b-a}\right) + b_k \sin\left(k \frac{2\pi t}{b-a}\right) \right), \\ a_k &= \frac{2}{b-a} \int_a^b f(t) \cos\left(k \frac{2\pi t}{b-a}\right) dt, \quad b_k = \frac{2}{b-a} \int_a^b f(t) \sin\left(k \frac{2\pi t}{b-a}\right) dt. \end{aligned}$$

- Complex form of the Fourier series

$$f(x) = \sum_{k=-\infty}^{\infty} c_k \exp(ikt), \quad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \exp(-kt) dt.$$

- Properties of the Fourier series coefficients

1. $\bar{c}_k = c_{-k}$
2. $a_k = a_{-k}, \quad b_{-k} = -b_k$
3. $f(t)$ even $\rightarrow \text{Im}(c_k) = 0, \quad f(t)$ odd $\rightarrow \text{Re}(c_k) = 0$
4. $b_0 = 0, \quad c_0 = \frac{a_0}{2}$

- Discrete Fourier transform (DFT): for a discrete signal $f[n]$ with $0 \leq n \leq N-1$

$$F[k] = \text{DFT} f[n] = \frac{1}{N} \sum_{n=0}^{N-1} f[n] W_N^{-kn}, \quad 0 \leq k \leq N-1, \quad W_N^k = \exp\left(i \frac{2\pi k}{N}\right).$$

- Properties of W_N

$$(W_N^k)^N = 1, \quad W_N^{-k} = W_N^{N-k}.$$

- Orthogonality relation on W_N

$$\sum_{k=0}^{N-1} W_N^{(i-j)k} = \begin{cases} N, & i = j \\ 0, & i \neq j \end{cases}$$

- Inverse discrete Fourier transform: Given a discrete frequency signal $F[k]$ with $0 \leq k \leq N-1$ then

$$f[n] = \text{IDFT} F[k] = \sum_{k=0}^{N-1} F[k] W_N^{kn}, \quad 0 \leq n \leq N-1.$$

- Properties of $F[k]$

1. $\text{Re}(F[k])$ is even in k
2. $\text{Im}(F[k])$ is odd in k
3. $\overline{F[k]} = F[-k]$
4. $f[n]$ is even in $n \rightarrow \text{Im}(F[k]) = 0$
5. $f[n]$ is odd in $n \rightarrow \text{Re}(F[k]) = 0$
6. $F[k] = F[k + sN]$ for $s \in \mathbb{Z}$

- FFT recursion: Given a discrete signal $f[n]$ $0 \leq n \leq N-1$ for $N = 2^m$ then the FFT algorithm uses the following recursion m times.

$$\begin{aligned} g[n] &= f[n] + f[n + N/2], \quad 0 \leq n \leq N/2 - 1 \\ h[n] &= (f[n] - f[n + N/2]) W_N^{-n} \quad 0 \leq n \leq N/2 - 1 \end{aligned}$$

where we can compute the DFT from

$$\begin{aligned} F[2l] &= \frac{1}{2} \text{DFT}\{g[n]\} \\ F[2l+1] &= \frac{1}{2} \text{DFT}\{h[n]\} \end{aligned}$$

- Euler's identity

$$\exp(ix) = \cos(x) + i \sin(x),$$

$$\exp(-ix) = \cos(x) - i \sin(x),$$

$$\cos(x) = \frac{\exp(ix) + \exp(-ix)}{2},$$

$$\sin(x) = \frac{(\exp(ix) - \exp(-ix))}{2i}$$